2014

Question 1

a)

1. $\beta_0 = 47.989, \beta_1 = 2.686$
2. $\hat{\sigma}^2 = 6.833^2 = 46.69$

b)

$$R^2 = 0.9192 = 91.92\%$$

c)

F-statistic's value is 79.64. With a p-value of 4.507×10^{-5}, and at a significance level of 5%, we have enough evidence to reject the null hypothesis. This indicates that there is a statistically significant linear relationship - and furthermore that the intercept explains a statistically significant amount of variation.

d)

The t-statistic's value is 8.924. To test a one-sided alternative, you would divide the p-value in half, as the t-distribution is symmetric.

e)

You can derive these yourselves, or it's contained in the lecture slides for week 2. To derive it yourself, CI for mean interval is about $\hat{y}(x_0)$, and the prediction interval is about $y(x_0) - \hat{y}(x_0)$.

$$\text{Prediction interval} = \hat{y}(x_0) \pm t_{\alpha/2, n-p}\hat{\sigma}\left(1 + \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}}\right)$$

$$\text{Confidence interval} = \hat{y}(x_0) \pm t_{\alpha/2, n-p}\hat{\sigma}\left(\frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}}\right)$$

Therefore, we have:

1. $101.79 \pm 2.364 \cdot 6.833 \cdot \left(1 + \frac{1}{9} + \frac{(20-21.37)^2}{463.24}\right) = (83.70, 119.72)$
2. $101.79 \pm 2.364 \cdot 6.833 \cdot \left(\frac{1}{9} + \frac{(20-21.37)^2}{463.24}\right) = (99.85, 103.57)$

Question 2

a)

We'd stop at `rings ~ length`, as at this step, theres no more predictors with a p-value < 0.05.

b)

1. γ is 46.401. You can find this by considering that `<none>` is just the base model of that step, and then look in the previous step, for what the `RSS` was there.

   δ is 23.676. Reminder that sequential sum of squares is the difference between $SS^{(2)}_{reg/res} - SS^{(1)}_{reg/res}$. You can find this by minusing `<none>`'s RSS to `data$weight`'s RSS in the null step.

2. A reminder that
$$F = \frac{R(\beta^{(2)}|R(\beta^1))}{SS_{reg}} \sim F_{1,n-r}$$
where r is the total number of parameters estimated in $\beta^{(2)}|\beta^{(1)}$. We have 14 samples, and in the first step for α, 2 parameters for each line being estimated (new parameter + intercept).

   So, we have:
$$\frac{\alpha}{51.538/(14-2)} = 5.5127$$
and thus that $\alpha = 23.68$.

   In a similar vein, β can be found by:
$$\beta = \frac{0.17801}{46.223/(14-4)} = 0.385$$
Just don't forget $p = k + 1$, where there are k predictors (+ intercept!)

3. So the models here are:
$$M_0 : y = \beta_0 + \beta_{length}$$
and
$$M_1 : y = \beta_0 + \beta_{length} + \beta_{diameter} + \beta_{weight}$$
This means we want to check whether $\beta_{diameter} = \beta_{weight} = 0$. We can check this by doing:
$$F = \frac{R(\beta_{diameter}, \beta_{weight}|\beta_{length})/2}{SS_{reg}/(14-4)} \sim F_{2,10}$$
The F-statistic equals:
$$\frac{R(\beta_{diameter}|\beta_{length}) + R(\beta_{weight}|\beta_{diameter}, \beta_{length})}{46.223/10} = \frac{2.309505}{4.6623} = 0.495$$
Since $0.495 < 4.102821$, **we cannot reject the null hypothesis**.

Question 3

a)

PRESS considers out of sample prediction, C_p considers subset selection bias-variance tradeoff, Adj. R^2 and R^2 both consider the total variation the regression component captures; where *adjusted* scales the regression component by the number of parameters used.

b)

$$R^2 = 1 - \frac{SS_{res}}{SS_{total}}$$

$$\bar{R}^2 = 1 - \frac{(n-1)SS_{res}}{(n-p)SS_{total}} = 1 - \frac{n-1}{n-p}(1-R^2)$$

Now whenever $\frac{n-1}{n-p}(1-R^2) > 1$, we have that $\bar{R}^2 < 0$.

Question 4

a)

$$\begin{aligned}
E(y^T A y) &= \sum_{i=1}^{n}\sum_{j=1}^{n} E(y_j A_{ji} y_i) \\
&= \sum_{i=1}^{n}\sum_{j=1}^{n} A_{ji} E(y_j y_i) \\
&= \sum_{i=1}^{n}\sum_{j=1}^{n} A_{ji}(V_{ij} + \mu\mu) \\
&= \sum_{i=1}^{n}\sum_{j=1}^{n} A_{ji} V_{ij} + \mu A_{ji} \mu \\
&= tr(AV) + \mu^T a \mu
\end{aligned}$$

b)

$$\begin{aligned}
E\left(\sum_{i=1}^{k} Z_i^2\right) &= \sum_{i=1}^{k} E(Z_i^2) \\
&= \sum_{i=1}^{k} \mathrm{Var}(Z_i) + E(Z_i)^2 \\
&= \sum_{i=1}^{k} \sigma^2 + \mu^2 \\
&= k(\sigma^2 + \mu^2)
\end{aligned}$$

$\sum_{i=1}^{k} Z_i^2 = \chi_k^2$, so the expectations are the same.

Question 5

I'm almost certain questions like c) and d) aren't asked anymore. If they ask us this... I mean, we've learnt 0 MGF stuff throughout the term, been asked almost 0 questions (besides 1 tute question), and have not used it in lectures.

a)

$$\begin{aligned}
\hat{\sigma}^2 &= \frac{1}{n-2}\sum_{i=1}^{n}(y_i - b_0 - b_1 x_i)^2 \\
&= \ldots (y_i - \bar{y} + b_1\bar{x} - b_1 x_i)^2 \\
&= \ldots (y_i - \bar{y} - b_1(x_i - \bar{x}))^2 \\
&= \ldots (y_i - \bar{y})^2 - 2b_1(x_i - \bar{x})(y_i - \bar{y}) + b_1^2(x_i - \bar{x})^2 \\
&= \frac{S_{yy} - 2b_1 S_{xy} + b_1^2 S_{xx}}{n-2} \\
&= \frac{S_{yy} - b_1^2 S_{xx}}{n-2} \qquad \text{Since } S_{xy} = b_1^2 S_{xx}
\end{aligned}$$

b)

$$\begin{aligned}
E(S_{yy}) &= \sum_{i=1}^{n} E[(y_i - \bar{y})^2] \\
&= \sum_{i=1}^{n} \text{Var}(y_i - \bar{y}) + \sum_{i=1}^{n} E(y_i - \bar{y})^2 \\
E(y_i - \bar{y}) &= E(\beta_0 + \beta_1 x_i + \epsilon_i - \beta_0 - \beta_1\bar{x}) \\
&= \beta_1(x_i - \bar{x}) \\
E(S_{yy}) &= \sum_{i=1}^{n} \text{Var}(y_i - \bar{y}) + \beta_1^2 S_{xx} \\
\text{Var}(y_i - \bar{y}) &= \text{Var}(y_i) + \text{Var}(\bar{y}) - 2\text{Cov}(y_i, \bar{y}) \\
&= \sigma^2 + \sigma^2/n - 2\sigma^2/n \\
&= \sigma^2 - \sigma^2/n \\
E(S_{yy}) &= n\sigma^2 - \sigma^2 + \beta_1^2 S_{xx}
\end{aligned}$$

Now considering the other component:

$$\begin{aligned}
E(b_1^2 S_{xx}) &= S_{xx} E(b_1^2) \\
&= S_{xx}(\text{Var}(b_1) + [E(b_1)]^2) \\
&= S_{xx}(\sigma^2/S_{xx} + \beta_1^2) \\
&= \sigma^2 + \beta_1^2 S_{xx}
\end{aligned}$$

This combines to $(n-2)\sigma^2$.

Question 6

a, b) are fairly uninteresting. These are pretty standard MLE derivations - this specific question can be found in one of the week's tutorials.

c)

Best, Linear and Unbiased Estimator. Best, alluding to lowest variance. Linear, and unbiased, which means it's expectation must be β.

d)

Consider an alternative estimator $\hat{b} = ((X^TX)^{-1}X^T + C)y$. It must unbiased so:

$$\begin{aligned} E(b + Cy) &= \beta + CX\beta \\ \beta + CX\beta &= \beta \qquad \text{[Since it must be unbiased]} \\ CX\beta &= 0 \\ CX &= 0 \end{aligned}$$

Now considering the variance:

$$\begin{aligned} \text{Var}(((X^TX)^{-1}X^T + C)y) &= ((X^TX)^{-1}X^T + C)\text{Var}(y)(X(X^TX)^{-1} + C^T) \\ &= \sigma^2((X^TX)^{-1} + (X^TX)^{-1}(CX)^T + CX(X^TX)^{-1} + CC^T) \\ &= \sigma^2((X^TX)^{-1} + CC^T) \qquad \text{[From the above derived identities]} \end{aligned}$$

Note $CC^T \sum_{i=1}^{n} \sum_{j=1}^{n} C_{ij}^2$, so $\text{Var}(\hat{b}) \geq \text{Var}(b)$. Thereby, BLUE.

Question 7

a)

$$\begin{pmatrix} y_1 \\ \dots \\ y_n \end{pmatrix} = \begin{pmatrix} 1 & x_1 \\ \dots & \dots \\ 1 & x_n \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} + \begin{pmatrix} \epsilon_1 \\ \dots \\ \epsilon_n \end{pmatrix}$$

b)

$$\begin{aligned} \text{Var}(b_1) &= \text{Var}\left(\frac{S_{xy}}{S_{xx}}\right) \\ &= \frac{1}{S_{xx}^2}\text{Var}\left(\sum_{i=1}(x_i - \bar{x})y_i\right) \\ &= \frac{\sigma^2}{S_{xx}} \\ \text{Var}(b_0) &= \frac{\sigma^2}{n} + \frac{\sigma^2}{S_{xx}} \end{aligned}$$

For the covariances, watch out for $\sum_{i=1}^{n}(x_i - \bar{x}) = 0$.

c)

$$\begin{aligned} \text{Var}(\hat{y}(x_0)) &= \text{Var}(b_0 + b_1x_0) \\ &= \text{Var}(b_0) + x_0^2\text{Var}(b_1) + 2x_0\text{Cov}(b_0, b_1) \\ &= \frac{\sigma^2}{n} + \frac{\sigma^2}{S_{xx}} + \frac{x_0^2\sigma^2}{S_{xx}} - \frac{2x_0\bar{x}\sigma^2}{S_{xx}} \\ &= \sigma^2\left(\frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}}\right) \end{aligned}$$

d)

Trick for this question is to recognise the variance formula from the last question. We want to use:

$$\text{Var}(x_i^T b) = \sigma^2 \left(\frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}} \right)$$

Now note the i-th entry for H can be represented as:

$$h_{ii} = x_i^T (X^T X)^{-1} x_i$$

where $x_i = (x_{i1}, x_{i2}, \ldots, x_{in})^T$. Note the flip of the transposes, due to this i-th row definition. Okay, now just:

$$\begin{aligned} \text{Var}(x_i^T b) &= x_i^T \text{Var}(b) x_i \\ &= \sigma^2 x_i^T (X^T X)^{-1} x_i \\ &= \sigma^2 h_{ii} \end{aligned}$$

Thereby completing the proof (compare coefficients).